

Diabetes Prediction Using Machine Learning Algorithms

Phase II – Machine Learning Project

Student: Ali "Mohammed Khair" Bani Domi | ID: 202120510

Project Overview

This project predicts diabetes using supervised machine learning techniques applied to the **Pima Indians Diabetes Dataset**, which contains medical diagnostic measurements for female patients of Pima Indian heritage.

The goal is to preprocess the data, train and compare multiple classification algorithms, and identify the best-performing model based on key evaluation metrics.

Project Structure

```
├─ Diabetes_Prediction_Phase_II.ipynb  # Main notebook
├─ diabetes.csv                        # Dataset (Pima Indians Diabetes)
└─ README.md
```

Dataset

- **Source:** Pima Indians Diabetes Dataset
 - **Features:** Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age
 - **Target:** `Outcome` (0 = No Diabetes, 1 = Diabetes)
-

Workflow

1. Data Preprocessing

- Replaced biologically invalid zero values in `Glucose`, `BloodPressure`, `SkinThickness`, `Insulin`, and `BMI` with `NaN`
- Applied **median imputation** for missing values
- Split data: **70% training** / **30% testing** (stratified)

- Applied **StandardScaler** and **MinMaxScaler** as appropriate per model

2. Models Trained

Model	Scaler Used
Logistic Regression	StandardScaler
K-Nearest Neighbors (KNN)	MinMaxScaler
Naive Bayes	StandardScaler
Support Vector Machine (SVM)	StandardScaler

3. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

How to Run

1. Clone the repository:

```
bash

git clone https://github.com/your-username/diabetes-prediction.git
cd diabetes-prediction
```

2. Install dependencies:

```
bash

pip install pandas numpy matplotlib seaborn scikit-learn jupyter
```

3. Launch the notebook:

```
bash
```

 Make sure `diabetes.csv` is in the same directory as the notebook.

Technologies Used

- Python 3.x
 - Pandas & NumPy
 - Matplotlib & Seaborn
 - Scikit-learn
-

Results

A comparison table of all four models is generated at the end of the notebook, evaluating performance across all metrics to identify the best classifier for diabetes prediction.

License

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